# MTH 202 Chapter 1: Three-Dimensional Space & Vectors

## 1.2 – 1.4 Vectors Worksheet

1. Let $\mathbf{u} = \langle 1,0,2 \rangle$, $\mathbf{v} = \langle -3,1,1 \rangle$, and $\mathbf{w} = \langle 2,-1,1 \rangle$ be vectors in $\mathbb{R}^3$.

a) Determine the exact value of $\|2\mathbf{u} - \mathbf{v}\|$.

b) Let $\theta$ be the angle between $\mathbf{u}$ and $\mathbf{v}$. Determine the exact value of $\theta$, then determine whether $0 < \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta < \pi$.

c) Mark the answer that best describes the meaning of the expression $|\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|$.

☐ It is the sum of the areas of the two parallelograms determined by the two pairs $\{\mathbf{u}, \mathbf{v}\}$ and $\{\mathbf{u}, \mathbf{w}\}$.

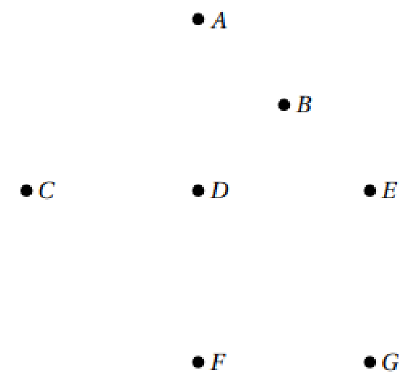
☐ It is the area of the parallelogram with vertices determined by 0, $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$.

☐ It is the volume of the parallelepiped determined by the vectors $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$.

☐ It has no meaning, but it is always defined and sometimes it is zero.

☐ It is undefined.

2. Consider the points in the plane shown at the right.




a) Determine the name of a vector that is equal to $2\overrightarrow{DE}-\overrightarrow{CA}$.

b) Determine the name of a vector that is equal to $\text{proj}_{\overrightarrow{CA}}\left(\overrightarrow{DE}\right)$.

3. Determine a unit vector which has the opposite direction of $\mathbf{w}=\langle 4,3,1\rangle$.

4. Consider the three points $A=(0,-1,1)$, $B=(1,-1,3)$, and $C=(2,0,0)$ in $\mathbb{R}^3$.

a) Find $\mathbf{v}=\overrightarrow{AB}$ and $\mathbf{w}=\overrightarrow{BC}$.

b) Calculate the cross-product $\mathbf{v}\times\mathbf{w}$.

c) Find the area of the parallelogram determined by **v** and **w**.

5. Determine whether the vectors $\mathbf{a}=\langle -2,5,3\rangle$ and $\mathbf{b}=\langle 2,2,-2\rangle$ are orthogonal, parallel, or neither.

6. Determine whether the vectors $\mathbf{a}=\langle -2,3,4\rangle$ and $\mathbf{b}=\langle 1,-3,2\rangle$ are orthogonal, parallel, or neither.

7. Consider the three points $A=(0,1,0)$, $B=(1,1,1)$, and $C=(0,2,1)$ in $\mathbb{R}^3$.

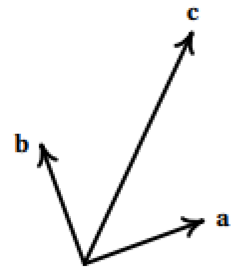
a) Determine the projection of the vector $\overrightarrow{AC}$ onto $\overrightarrow{AB}$.

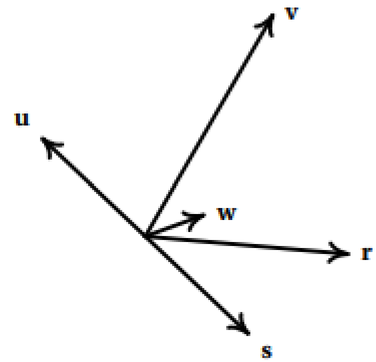
b) Determine the area of the triangle formed by these three points.

c) Calculate $\overrightarrow{AB}\cdot\left(\overrightarrow{AB}\times\overrightarrow{AC}\right)$.

d) For the vector $\mathbf{v}=\langle 0,2,2\rangle$, determine the angle between $\overrightarrow{AC}$ and $\mathbf{v}$.

8. Consider the vectors **a**, **b**, and **c** shown at the right. Their lengths are $\|\mathbf{a}\| = \|\mathbf{b}\| = 1$ and $\|\mathbf{c}\| = 2$.




a) Circle the vector that best represents $\frac{1}{2}\mathbf{a} - \mathbf{b}$.

| | | | | |
|---|---|---|---|---|
| **u** | **v** | **w** | **r** | **s** |

b) Circle the number closest to $\mathbf{a} \cdot \mathbf{c}$.

| | | | | |
|---|---|---|---|---|
| 0 | 1 | $\sqrt{2}$ | $\sqrt{3}$ | 2 |

c) A nonzero vector **n** is pointing directly up out of the page. Circle the best description of $\mathbf{b} \times \mathbf{c}$.

| | | |
|---|---|---|
| it is a positive multiple of **n** | it is a negative multiple of **n** | it is not a multiple of **n** |

9. Determine all values of $a$ for which the vectors $\langle a, -3, 1 \rangle$ and $\langle a, 2a, -7 \rangle$ are perpendicular.

10. Let **a** and **b** be two vectors in $\mathbb{R}^3$ such that $\left\|\text{proj}_{\mathbf{a}}(\mathbf{b})\right\| = 2$.

a) Determine the value of $\left\|\text{proj}_{3\mathbf{a}}(\mathbf{b})\right\|$.

b) Determine the value of $\left\|\text{proj}_{\mathbf{a}}(5\mathbf{b})\right\|$.

11. Determine all values of $a$ and $b$ for which $\langle a,-3,2b\rangle$ and $\langle a,a,1\rangle$ are parallel.